

DOKUMENTACJA GAIA

1. Opis i Wymagania

Proponowany system organizacji i informowania o transporcie w sytuacjach kryzysowych, takich jak powódź, ma na celu zapewnienie skutecznego i sprawnego zarządzania transportem oraz komunikacją w obliczu zagrożeń. System ten składa się z kilku kluczowych komponentów, które współpracują ze sobą, aby minimalizować skutki kryzysów i wspierać mieszkańców.

- Zgłaszanie potrzeby zaopatrzenia o konkretne zasoby Zgłaszanie chęci zapewnienia konkretnych zasobów
- Dopasowywanie dostępnych zasobów do potrzeb Wyznaczanie optymalnych tras przejazdu Wyznaczanie alternatywnych tras przejazdu w razie wypadków losowych
- Aktualizacja obecnej trasy w razie wypadków losowych Zgłaszanie informacji o nieprzejezdnych drogach Wysyłanie powiadomień dotyczących zagrożeń
- Wysyłanie powiadomień dotyczących zmian w trasach transportowych
- Wysyłanie specjalnych powiadomień dla służb porządkowych

2. Model Ról

Role: ResourceRequest

Description	Represents a formal request for specific resources by a user or entity in a crisis situation. Contains details about required items, quantities, and urgency level.
Responsibilities	Liveness: SubmitResourceRequest^ω Safety:
Permissions	READ: WRITE: ResourceRequest CONSUME: PRODUCE:
Activities	
Protocols	SubmitResourceRequest

Role: ResourceOffer

Description	Represents an offer to provide specific resources by a user or entity. Contains details about available items, quantities, and location of availability.
Responsibilities	Liveness: SubmitResourceOffer^ω Safety:
Permissions	READ: WRITE: ResourceOffer CONSUME: PRODUCE:
Activities	
Protocols	SubmitResourceOffer

Role: ResourceMatch

Description	Represents the result of matching available resources with resource requests. Contains matched pairs of requests and offers along with allocation details.
Responsibilities	Liveness: MatchResources . ProcessResourceMatch Safety: UniqueID(ResourceMatch) != Null ResourceMatch.RequestID != Null ResourceMatch.OfferID != Null
Permissions	READ: ResourceRequest, ResourceOffer WRITE: ResourceMatch CONSUME: ResourceRequest, ResourceOffer PRODUCE: ResourceMatch
Activities	ProcessResourceMatch
Protocols	MatchResources

Role: TransportRoute

Description	Represents a planned or actual transport route including waypoints, estimated travel time, and current status. Used for planning and real-time updates.
Responsibilities	Liveness: PlanTransportRoute . PlanRoute UpdateTransportRoute . UpdateRouteStatus Safety: UniqueID(TransportRoute) != Null TransportRoute.Status != Null TransportRoute.Waypoints != Null
Permissions	READ: TransportRoute WRITE: TransportRoute CONSUME: TransportRoute PRODUCE: TransportRoute
Activities	PlanRoute, UpdateRouteStatus
Protocols	PlanTransportRoute, UpdateTransportRoute

Role: RouteUpdate	
Description	Represents an update to an existing transport route due to unforeseen circumstances such as accidents or road closures. Contains new route information and reasons for changes.
Responsibilities	Liveness: UpdateTransportRoute . ProcessRouteUpdate ReportRoadBlockage . ProcessRouteUpdate Safety: UniqueID(RouteUpdate) != Null RouteUpdate.Reason != Null RouteUpdate.NewRoute != Null
Permissions	READ: RouteUpdate, RoadBlockageReport WRITE: RouteUpdate CONSUME: RouteUpdate, RoadBlockageReport PRODUCE: RouteUpdate
Activities	ProcessRouteUpdate
Protocols	UpdateTransportRoute, ReportRoadBlockage

Role: RoadBlockageReport	
Description	Represents a report of a road blockage or impassable section of road. Contains location details, severity, and estimated duration of the blockage.
Responsibilities	Liveness: ReportRoadBlockage^ω Safety: UniqueID(RoadBlockageReport) != Null RoadBlockageReport.Location != Null RoadBlockageReport.Severity != Null
Permissions	READ: WRITE: RoadBlockageReport CONSUME: PRODUCE:
Activities	
Protocols	ReportRoadBlockage

Role: EmergencyNotification	
Description	Represents a notification sent to users regarding emergency situations, including warnings about road conditions, changes in transport routes, or other critical information.
Responsibilities	Liveness: SendEmergencyNotification . SendNotification Safety: UniqueID(EmergencyNotification) != Null EmergencyNotification.Message != Null EmergencyNotification.Recipients != Null
Permissions	READ: EmergencyNotification WRITE: EmergencyNotification CONSUME: EmergencyNotification PRODUCE: EmergencyNotification
Activities	SendNotification
Protocols	SendEmergencyNotification

Role: SpecializedNotification

Description	Represents a specialized notification sent to emergency services or specific authorities. Contains critical information relevant to law enforcement, rescue teams, or other emergency responders.
Responsibilities	Liveness: SendSpecializedNotification . SendSpecializedNotification Safety: UniqueID(SpecializedNotification) != Null SpecializedNotification.CriticalInfo != Null SpecializedNotification.TargetAuthorities != Null
Permissions	READ: SpecializedNotification WRITE: SpecializedNotification CONSUME: SpecializedNotification PRODUCE: EmergencyNotification
Activities	SendSpecializedNotification
Protocols	SendSpecializedNotification

3. Model Interakcji

Goal	Name	Initiator	Responder	Inputs	Outputs	Description
Resource	SubmitResourceRequest	ResourceRequest	ResourceMatch	ResourceRequest		Initiates the process of submitting a request for specific resources during a crisis.
Resource	SubmitResourceOffer	ResourceOffer	ResourceMatch	ResourceOffer		Initiates the process of submitting an offer to provide specific resources during a crisis.
Available with needs	MatchResources	ResourceMatch	ResourceRequest	ResourceRequest, ResourceOffer	ResourceMatch	Processes submitted resource requests and offers to find optimal matches for allocation.
Transport route	PlanTransportRoute	TransportRoute	RouteUpdate	TransportRoute	TransportRoute	Initiates the process of planning a transport route based on current conditions.
Transport	UpdateTransportRoute	RouteUpdate	TransportRoute	RouteUpdate	TransportRoute	Updates an existing transport route due to unforeseen circumstances such as accidents or road closures.
Road	ReportRoadBlockage	RoadBlockageReport	RouteUpdate	RoadBlockageReport	RouteUpdate	Reports a road blockage or impassable section of road to the system about current road conditions.
Emergency	SendEmergencyNotification	EmergencyNotification	ResourceRequest	EmergencyNotification		Sends a notification to users about emergency situations or changes in transport routes.
Specialized	SendSpecializedNotification	SpecializedNotification	EmergencyNotification	SpecializedNotification	EmergencyNotification	Sends a specialized notification about emergency services or specialized authorities.

4. Model Usage

Service	Agent	In/Out	Pre/Post Condition	Derived From
SubmitResourceRequest	ResourceCoordinatorAgent	IN: ResourceRequest OUT:	PRE: ResourceRequest is valid and contains required fields POST: ResourceRequest is submitted and stored in the system	SubmitResourceRequest
SubmitResourceOffer	ResourceCoordinatorAgent	IN: ResourceOffer OUT:	PRE: ResourceOffer is valid and contains required fields POST: ResourceOffer is submitted and stored in the system	SubmitResourceOffer
MatchResources	ResourceCoordinatorAgent	IN: ResourceRequest, ResourceOffer OUT: ResourceMatch	PRE: ResourceRequest and ResourceOffer are valid and exist in the system POST: ResourceMatch is created with unique ID, RequestID and OfferID are not null	MatchResources
ProcessResourceMatch	ResourceCoordinatorAgent	IN: ResourceRequest, ResourceOffer OUT: ResourceMatch	PRE: ResourceRequest and ResourceOffer are valid and exist in the system, ResourceMatch does not exist POST: ResourceMatch is created with unique ID, RequestID and OfferID are not null, ResourceMatch is stored in the system	ProcessResourceMatch
PlanTransportRoute	TransportPlannerAgent	IN: TransportRoute OUT: TransportRoute	PRE: TransportRoute is valid and contains required fields POST: TransportRoute is planned with unique ID, Status and Waypoints are not null	PlanTransportRoute
UpdateTransportRoute	TransportPlannerAgent	IN: RouteUpdate OUT: TransportRoute	PRE: RouteUpdate is valid and exists in the system, TransportRoute exists and is valid POST: TransportRoute is updated with new route information and reason for changes, RouteUpdate is processed	UpdateTransportRoute
ProcessRouteUpdate	TransportPlannerAgent	IN: RouteUpdate OUT: TransportRoute	PRE: RouteUpdate is valid and exists in the system, TransportRoute exists and is valid POST: TransportRoute is updated with new route information and reason for changes, RouteUpdate is processed	ProcessRouteUpdate
ReportRoadBlockage	TransportPlannerAgent	IN: RoadBlockageReport OUT: RouteUpdate	PRE: RoadBlockageReport is valid and registered in the system POST: RouteUpdate is created with unique ID, Reason and NewRoute are not null, RoadBlockageReport is processed	ReportRoadBlockage
SendEmergencyNotification	NotificationAgent	IN: EmergencyNotification OUT:	PRE: EmergencyNotification is valid and contains required fields POST: EmergencyNotification is sent to recipients, UniqueID and Message are not null	SendEmergencyNotification
SendSpecializedNotification	NotificationAgent	IN: SpecializedNotification OUT: EmergencyNotification	PRE: SpecializedNotification is valid and contains required fields POST: EmergencyNotification is created with unique ID, CriticalInfo and TargetAuthorities are not null, EmergencyNotification is stored in the system	SendSpecializedNotification

5. Model Znajomości

TransportPlannerAgent



ResourceCoordinatorAgent



NotificationAgent

